Semih Yagli (U.S. Lawful Permanent Resident)

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GitHub: github.com/semihyagli   ResearchGate: researchgate.net/profile/Semih-Yagli

Data Scientist with expertise in data analysis, machine learning, quantitative research and development, and full-stack application development, passionate about delivering creative, data-driven solutions to complex problems and generating actionable insights.

Educational Background

Princeton University, Princeton, New Jersey Sep. 2016 – May 2021

Electrical and Computer Engineering

Ph.D. in Electrical and Computer Engineering Dissertation Advisor: Prof. H. Vincent Poor

M.A. in Electrical Engineering GPA: 3.98/4.00

Orta Doğu Technical University, Ankara, Turkey (Double-Major) Sep. 2009 – May 2014

Mathematics, **Bachelor of Science**, GPA: 3.96/4.00

Electrical and Electronics Engineering, **Bachelor of Science**, GPA: 3.96/4.00

Patents

Machine Learning-Based Item Reranking Based on User Query and Cart Context	Patent No.: US20250342516A1	2025
Method and System for Fulfillment-Aware Product Search Ranking	(under review)	2026
Automatic Configuration Management in Cellular Networks	(under review)	2022

Coding and Technical

Programming Languages:	Python (XGBoost, Keras, PyTorch, sklearn), Java, JavaScript, React, React Native, bash.
Data & Analytics:	SQL, NoSQL, BigQuery, R, Python (Pandas, PySpark, Matplotlib), Tableau.
Machine Learning & AI:	Neural Networks, Large Language Models, Natural Language Processing (NLP), Classification, Bayesian Inference, Clustering, Regression Analysis.
Development & Platforms:	Google Cloud (GCP), BigQuery, Firestore, Azure, Amazon Web Services, Axios HTTP.
Quantitative Skills:	Probability Theory, Optimization, Data Visualization.

Work Experience

Walmart Global Tech, Staff Data Scientist Oct. 2021 – Present

Lead end-to-end machine learning (ML) model development for e-commerce Search and BuyBox ranking systems, integrating engineering and data science capabilities across Python and Java. Conduct advanced experimentation, signal design, model training, and runtime deployment to drive large-scale business outcomes. Inform strategy through rigorous A/B testing, support infrastructure improvements, and mentor junior data scientists on navigating Walmart's complex systems.

Senior Data Analytics & Machine Learning Engineer at Search Ranking:

- **Delivered a 2.4% GMV lift** by architecting a unified ranking system that harmonized default and personalized search rankers, replacing fragmented legacy models with a high-performance, scalable framework.
- **Improved query-item relevance by 11%** by engineering a novel token-importance signal; spearheaded the end-to-end integration via **Java-based CI/CD** pipelines and advanced ML model deployments.
- **Directed cross-functional model validation** using interleaving and A/B testing frameworks, ensuring statistical rigor and driving consensus among senior stakeholders for high-stakes production changes.
- **Optimized system reliability and latency** by deploying high-throughput ranking features within core infrastructure.

Staff Data Scientist at Buy Box Offer Ranking:

- **Spearheaded Product Category feature development**, driving a site-wide **0.87% lift** in Multi-Offer Add-To-Cart (ATC) rates, generating an estimated **$58M in incremental annual revenue**.
- **Increased ATC conversions by 4%** by architecting a custom **Constrained Neural Network** to enforce price and shipping monotonicity, ensuring high-precision ranking logic.
- **Enhanced multi-offer conversion rates by 3.8%** by modeling multi-dimensional relationships between shipping speed, pricing, and buyer intent to optimize item-page rankings.
- **Automated end-to-end model training and deployment pipelines**, significantly reducing iteration time and enabling seamless retraining and rapid experimentation cycles.

Technologies: Java (Runtime & Unit Testing), Python (PyTorch, Keras, Pandas, XGBoost), Google Cloud (GCP), BigQuery, Azure, Tableau, SQL.

Personal Projects and Experiences

Portfolio Optimization & Mid-Frequency Trading May 2025 – Present

- **Spearheaded** end-to-end design of systematic trading strategies spanning portfolio optimization and mid-frequency trading, with long-term decision support driven by signals extracted from SEC filings.
- **Engineered** an Adaptive Kalman Filter-based allocation model, achieving a **4–6% gain** in information coefficient and an **8% reduction** in portfolio volatility across multiple market-regime backtests.

Technologies: Python (numpy, pandas, yfinance, financetoolkit, financedatabase), Interactive Brokers Paper Trading.

ACTUS / FINOS CDM Collaboration 2025 – Present

- **Leading** integration work extending the FINOS Common Domain Model (CDM) to natively support ACTUS cashflow calculations, bridging the gap between financial contract standards and open-source infrastructure.
- Driving design and implementation in collaboration with ACTUS-affiliated and independent researchers, building a reusable framework for standardized financial instrument modeling.

Machine Learning Competitions 2025 – 2026

- **Competing** in the KDD Cup 2025 Ad Recommendation Challenge, building large-scale recommendation models and ranking systems informed by industry-grade ML practices.
- **Targeting** the RecSys Challenge 2026, applying deep expertise in ranking, personalization, and collaborative filtering to a competitive research setting.

Etiqueta – AI Data Labeling Platform, Founder & Lead Developer Mar. 2023 – May 2025

Architected and deployed **Etiqueta**, a first-of-its-kind mobile-native platform for AI-aided data labeling and image segmentation. Engineered the full technology stack to enable scalable, gamified crowdsourcing across iOS, Android, and ARM-based desktop environments.

Platform Impact & Open Source Contribution

- **Curated 100K+ high-fidelity labels** and **13K image segmentations** via gamified workflows; published open-source datasets on Kaggle and Hugging Face (1K+ downloads).
- **Published a technical report** and codebase on GitHub, establishing a framework for mobile-first dataset distribution and research adoption.
- **Launched on Global App Stores** as the only public, mobile-native data labeling solution, managing the full product lifecycle from UX design to store optimization.

Full-Stack Engineering

- **Frontend:** Developed a high-throughput React Native interface supporting 30+ concurrent label interactions per screen, optimizing UI responsiveness and reducing interaction latency.
- **Backend:** Built a scalable Python (FastAPI) and Firestore architecture to handle real-time task distribution, utilizing OAuth for secure authentication and end-to-end data privacy.
- **Reliability:** Implemented comprehensive backend unit testing (pytest) and CI/CD practices to ensure cross-platform stability across iOS, Android, and macOS.

Technologies: Python (FastAPI), React Native, JavaScript, Firestore (NoSQL), OAuth, Axios, Pytest.

Awards and Honors

Excellence in Teaching Award at Princeton University 2018
Turkish Presidential Fellowship Award 2009 – 2014
Ranked 78th out of 1.5M+ candidates in the Turkish National University Entrance Exam 2008

Extra Curricular

Languages: Fluent in English, French, Spanish, and Turkish (Native).
Mentorship: Volunteer mentor at Princeton GradFutures program; each year mentoring about 4 students.
Leadership: Regional Princeton Graduate Alumni Leader; organizing networking events for alumni.

Ph.D. research highlights and full publication list available upon request.

Ph.D. Research Highlights

Bounded Gaussian Mean Estimation, 50-year-old open problem

- Used non-linear optimization to prove that the least favorable distribution has finite support.
- My technique based on the oscillation theorem has been extensively used in multiple-access and wiretap channel setups.

Information Secrecy and User Privacy

- Developed a statistical method, called poissonization, to simplify the analysis of weakly dependent random variables.
- My 'poissonization' proof technique has been employed in large deviations and quantum information theory research.

Privacy in Machine Learning

- Used supervised and adversarial machine learning tools to remove sensitive information from data.
- Ensured minimal computational burden on the user to make the solution feasible for all types of ML prediction tasks.

Professional Services

Technical Program Committees:

IEEE 54th Annual Conference on Information Sciences and Systems (CISS 2020), Princeton, NJ, USA, March 18–20, 2020.
IEEE 52nd Annual Conference on Information Sciences and Systems (CISS 2018), Princeton, NJ, USA, March 22–23, 2018.

Publications

Ongoing Work:

S. Yagli, "Etiqueta: AI-Aided, Gamified Data Labeling to Label and Segment Data," *Manuscript prepared.*

A. Dytso, S. Yagli, and L. Barletta (Eds.), "Collected Works of Hans Sylvain Withenhausen," *Manuscript prepared.*

Journal Papers:

M. Al, S. Yagli, and S. Y. Kung, "Privacy Enhancing Machine Learning via Removal of Unwanted Dependencies," *IEEE Transactions on Neural Networks and Learning Systems*, Sep. 2021.

A. Dytso, S. Yagli, H. V. Poor, and S. Shamai (Shitz), "Capacity Achieving Distribution for the Amplitude Constrained Additive Gaussian Channel: An Upper Bound on the Number of Mass Points," *IEEE Transactions on Information Theory*, vol. 66, no. 4, pp. 2006–2022, Apr. 2020.

S. Yagli and P. Cuff, "Exact Exponent for Soft Covering," *IEEE Transactions on Information Theory*, vol. 65, no. 10, pp. 6234-6262, Oct. 2019.

S. Yagli, Y. Altuğ, and S. Verdú, "Minimax Rényi Redundancy," *IEEE Transactions on Information Theory*, vol. 64, no. 5, pp. 3715–3733, May 2018.

Conference Proceedings:

S. Yagli, A. Dytso, and H. V. Poor, "Information-Theoretic Bounds on the Generalization Error and Privacy Leakage in Federated Learning," in *Proc. 21st IEEE International Workshop On Signal Processing Advances in Wireless Communications (SPAWC 2020)*, Atlanta, GA, USA, May 2020.

S. Yagli, A. Dytso, and H. V. Poor, "Estimation of Bounded Normal Mean: An Alternative Proof for the Discreteness of the Least Favorable Prior," in *Proc. 2019 IEEE Information Theory Workshop (ITW)*, Visby, Sweden, Aug. 2019.

S. Yagli, A. Dytso, H. V. Poor, and S. Shamai (Shitz), "An Upper Bound on the Number of Mass Points in the Capacity Achieving Distribution for the Amplitude Constrained Additive Gaussian Channel," in *Proc. 2019 IEEE International Symposium on Information Theory (ISIT)*, Paris, France, Jul. 2019, pp. 1907–1911.

S. Yagli, A. Dytso, H. V. Poor, and S. Shamai (Shitz), "Some Aspects of Totally Positive Kernels Useful in Information Theory," in *Proc. 2019 MoTION Workshop of Wireless Communications and Networking Conference (WCNC)*, Marrakech, Morocco, Apr. 2019, pp. 1–6.

S. Yagli and P. Cuff, "Exact Soft-Covering Exponent," in *Proc. 2018 IEEE International Symposium on Information Theory (ISIT)*, Vail, CO, Jun. 2018, pp. 1680–1684.

S. Yagli, Y. Altuğ, and S. Verdú, "Minimax Rényi Redundancy," in *Proc. 2017 IEEE International Symposium on Information Theory (ISIT)*, Aachen, Germany, Jun. 2017, pp. 2980–2984.

Dissertation:

S. Yagli, "Topics in Information and Estimation Theory: Parameter Estimation, Lossless Compression, Constrained Channels, and Error Exponents," PhD Dissertation, Princeton University, 2021.

Technical Reports:

S. Temel, S. Yagli, and S. Gören, "P, PD, PI, PID Controllers," Middle East Technical University, Ankara, Turkey, Technical Report, 2013.